

Operations Research

Lecture 5: Simplex Tableau and Sensitivity Analysis

A textbook treatment

Based on the lecture notes of El Mehdi Saad

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How to read this book

Each concept in this book is presented in the same rhythm.

- 1. Motivation and intuition.** Why do we care? What problem does the idea solve, and what picture should you hold in your head?
- 2. Formal statement.** The precise definition or theorem, set in a coloured box so it is easy to find when revising.
- 3. Proof.** Every claim is proved. Longer proofs are split into labelled steps, and the one idea that makes the proof work is isolated in a *Key insight* box.
- 4. Interpretation and discussion.** What the result really says, where it applies, and which misreadings to avoid.
- 5. Examples.** From “trivially small” to “genuinely useful.”
- 6. Techniques, classic problems and gotchas.** Computational shortcuts, proof patterns, exam favourites with one-line solutions, and the mistakes everyone makes once.

Margin notes carry the informal, “say it out loud” version of what is happening in the main text. Every major section closes with a summary box.

Prerequisites. Comfort with vectors, matrices, systems of linear equations, row reduction and the notions of rank and linear independence. Chapter 2 recalls exactly what is needed as it goes.

Chapter 5

The Simplex Tableau in Matrix Form and Sensitivity Analysis

5.1 A Simplex Refresher, and the Plan of the Chapter

Motivation

Chapter 3 ran the simplex method on tableaux, and Chapter 4 showed that the final objective row hides a dual solution. This chapter opens the tableau up and proves everything inside it. Part one rewrites a simplex move in matrix language: a *basic direction*, its *step length* and its *gain*, the reduced cost. Out of this come the optimality theorem, the ratio test, the basis exchange and termination, each with a complete proof. Part two uses the same matrices for *sensitivity analysis*: when a datum of an already-solved LP changes, does the old optimal basis survive, what is the new value, and how do we continue from the last tableau instead of starting over?

The tableau is not just bookkeeping. Every number in it is a piece of B^{-1} , and the whole chapter is about reading those numbers and proving why they mean what they mean.

This chapter is self-contained. Every result it uses is stated and proved here, including those first proved in earlier chapters:

Result	First seen	Proved here as
The tableau of a basis; row operations keep solutions	Ch. 2, 3	Theorem 5.2.2
Ratio test and unboundedness	Ch. 3	Theorem 5.3.3
A pivot gives a new BFS	Ch. 3	Theorem 5.3.4
Optimality test and uniqueness	Ch. 3	Theorem 5.4.2, Corollary 5.4.3
Termination without degeneracy	Ch. 3	Theorem 5.4.5
An optimal BFS exists	Ch. 2	Theorem 5.4.8
Weak duality, matching pair	Ch. 4	Theorems 5.5.1, 5.5.2
Simplex gives the dual solution	Ch. 4	Theorem 5.5.3

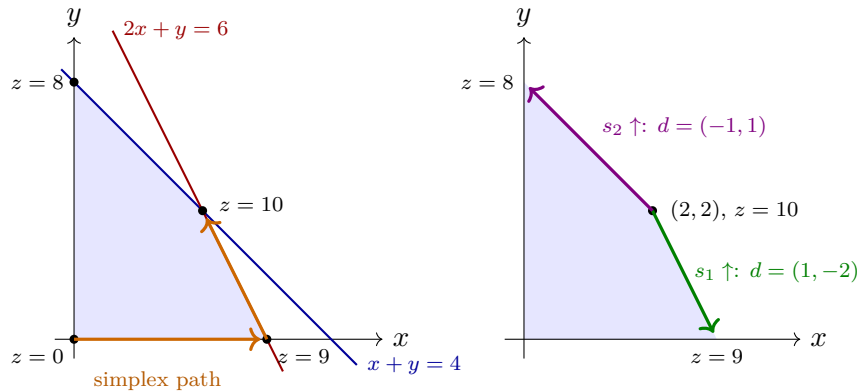
Example 5.1.1 (The running example). Two divisible activities $x, y \geq 0$ earn 3 and 2 per unit and share two resources:

$$\max 3x + 2y \quad \text{s.t.} \quad x + y \leq 4, \quad 2x + y \leq 6, \quad x, y \geq 0.$$

With slacks $s_1, s_2 \geq 0$, simplex runs as in Chapter 3. The slides call the objective z , where Chapter 3 wrote v .

Tableau 0 (basis s_1, s_2)	Tableau 1 (basis x, s_1)	Tableau 2 (basis x, y)
$s_1 = 4 - x - y$	$x = 3 - \frac{1}{2}y - \frac{1}{2}s_2$	$x = 2 + s_1 - s_2$
$s_2 = 6 - 2x - y$	$s_1 = 1 - \frac{1}{2}y + \frac{1}{2}s_2$	$y = 2 - 2s_1 + s_2$
$z = 0 + 3x + 2y$	$z = 9 + \frac{1}{2}y - \frac{3}{2}s_2$	$z = 10 - s_1 - s_2$
x in; ratios 4, 3; s_2 out	y in; ratios 6, 2; s_1 out	optimal: $z \leq 10$

The path is $(0, 0) \rightarrow (3, 0) \rightarrow (2, 2)$, with values $0 \rightarrow 9 \rightarrow 10$. In the last tableau every feasible plan has $s_1, s_2 \geq 0$, so $z = 10 - s_1 - s_2 \leq 10$, and the plan $(2, 2)$ is optimal.



The right-hand picture shows the two *basic directions* at the optimum, both downhill. Section 5.3 explains where they come from.

5.2 Bases and the Tableau of a Basis

The standing setting

Throughout, the LP is in equational form:

$$\max c^\top x \quad \text{s.t.} \quad Ax = b, \quad x \geq 0, \quad P = \{x \in \mathbb{R}^n : Ax = b, \quad x \geq 0\},$$

where $A \in \mathbb{R}^{m \times n}$ has **full row rank** m . Redundant equations can always be deleted first (Chapter 2). For an index set S , x_S , c_S and A_S denote the entries of x and c , and the columns of A , with indices in S .

Definition 5.2.1 (Basis, BFS, nondegeneracy). Choose m column indices $B(1), \dots, B(m)$ whose columns form an invertible matrix $B = [A_{B(1)} \cdots A_{B(m)}]$ (a *basis*), and let N be the other, *nonbasic*, indices. The *basic solution* of B sets $x_i = 0$ for $i \in N$ and $x_B = B^{-1}b$. If $x_B \geq 0$ it is a *basic feasible solution* (BFS). It is *nondegenerate* if every component of x_B is strictly positive.

The slides use the same letter B for the list of indices and for the matrix; the context always tells which.

Every solution of the equations, written in the basis

The simplex tableau of Chapter 3 was built by pivots. The next theorem shows that it is really a formula in B^{-1} , and that it does not depend on the pivots used to reach it.

Once you fix the nonbasic variables, the basic ones are forced. That single fact is the whole content of a tableau.

Theorem 5.2.2 (The tableau of a basis). *Let B be a basis and define the reduced costs $r^\top := c^\top - c_B^\top B^{-1}A$. Then:*

- (a) *For every $x \in \mathbb{R}^n$: $Ax = b$ if and only if $x_B = B^{-1}b - B^{-1}A_N x_N$. In particular, x_N can be chosen freely, and x_B is then determined.*
- (b) *For every x with $Ax = b$: $c^\top x = c_B^\top B^{-1}b + \sum_{j \in N} r_j x_j$.*
- (c) *$r_j = 0$ for every basic index j .*
- (d) **Uniqueness.** *Suppose that, for some vector β , matrix Q , number z_0 and vector ρ , every solution of $Ax = b$ satisfies $x_B = \beta + Qx_N$ and $c^\top x = z_0 + \rho^\top x_N$. Then $\beta = B^{-1}b$, $Q = -B^{-1}A_N$, $z_0 = c_B^\top B^{-1}b$ and $\rho = r_N$.*

Proof. (a) Split the product by columns: $Ax = Bx_B + A_N x_N$. So $Ax = b$ is the same equation as $Bx_B = b - A_N x_N$. Because B is invertible, this holds exactly when $x_B = B^{-1}(b - A_N x_N)$.

(b) Split the objective the same way and substitute (a):

$$\begin{aligned} c^\top x &= c_B^\top x_B + c_N^\top x_N = c_B^\top B^{-1}b - c_B^\top B^{-1}A_N x_N + c_N^\top x_N \\ &= c_B^\top B^{-1}b + (c_N^\top - c_B^\top B^{-1}A_N)x_N. \end{aligned}$$

The j -th entry of the bracket is $c_j - c_B^\top B^{-1}A_j = r_j$.

(c) On the basic columns, $r_B^\top = c_B^\top - c_B^\top B^{-1}B = c_B^\top - c_B^\top = 0$.

(d) By (a), for every choice of x_N there is a solution of $Ax = b$. Take $x_N = 0$. Then (a) gives $x_B = B^{-1}b$, the hypothesis gives $x_B = \beta$, and the objective gives $c_B^\top B^{-1}b = z_0$. Next take $x_N = e_j$ (one nonbasic variable equal to 1, the others 0). Comparing the two formulas for x_B gives $\beta + Q_{\cdot j} = B^{-1}b - B^{-1}A_j$, so $Q_{\cdot j} = -B^{-1}A_j$. Comparing the two formulas for $c^\top x$ gives $\rho_j = r_j$. $\square$

Interpretation & discussion

- **Why a pivoted tableau equals this formula.** Each pivot of Chapter 3 solves one equation for one variable and substitutes the result into the other equations and the objective. Each such step is reversible (substitute back), so the system after the pivot has exactly the same solutions as before. The final tableau therefore holds for *every* solution of $Ax = b$, and by (d) its numbers must be the ones in the theorem. The tableau of a basis depends only on the basis, not on the path that reached it.
- **The identity in (b) holds for every solution of $Ax = b$,** feasible or not. This is what makes the objective row a proof of optimality, and it is used in almost every proof below.

Corollary 5.2.3 (What the tableau displays). *Write the tableau of B as $x_{B(k)} = \beta_k + \sum_{j \in N} q_{kj} x_j$ for $k = 1, \dots, m$, and $z = z_0 + \sum_{j \in N} r_j x_j$. Then:*

- (i) *the constants are $\beta = B^{-1}b = x_B$, and $z_0 = c_B^\top B^{-1}b$ is the value of the basic solution;*
- (ii) *the column of a nonbasic x_j in the basic rows is $q_{\cdot j} = -B^{-1}A_j$;*
- (iii) *the objective coefficients are the reduced costs r_j ;*
- (iv) *if the original problem has a slack s_i with column e_i and cost 0, then its tableau column is $-B^{-1}e_i$ (minus the i -th column of B^{-1}), and its*

reduced cost is $-p_i$, where $p^\top := c_B^\top B^{-1}$. When all slacks are nonbasic, the slack columns together display $-B^{-1}$.

Proof. (i)–(iii) are Theorem 5.2.2(d). For (iv), apply (ii) with $A_j = e_i$, and compute $r_{s_i} = 0 - c_B^\top B^{-1}e_i = -p_i$. $\square$

Example 5.2.4 (B^{-1} in the running example). In Tableau 2 the slack coefficients are, row by row, $(1, -1)$ for x and $(-2, 1)$ for y . Hence

$$-B^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix} \implies B^{-1} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix},$$

for $B = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$, the columns of x and y . Check: $B^{-1}b = (-4 + 6, 8 - 6) = (2, 2)$, the optimal plan. The slack reduced costs are $-1, -1$, so $p = (1, 1)$.

Gotcha

- **The sign.** Because the basic variables are isolated on the *left*, the tableau shows $-B^{-1}$, not B^{-1} , and $-p$, not p . Forgetting these minus signs is the most common error in this chapter.
- **Notation clash.** Chapter 3 called the tableau constants p_i . Here, following the slides, p is the *multiplier vector* $c_B^\top B^{-1}$, and the constants are β_k .

5.3 Moving Between Bases: Directions, Ratio Test, Exchange

Definition 5.3.1 (Feasible direction). A vector d is a *feasible direction* at $x \in P$ if $x + \theta d \in P$ for some $\theta > 0$.

One move: raise one nonbasic variable

At a BFS, pick a nonbasic $j \in N$ and let x_j rise from 0 to $\theta > 0$, keeping the other nonbasic variables at zero. So $d_j = 1$ and $d_i = 0$ for $i \in N \setminus \{j\}$. The basic variables must adjust, $x_B(\theta) = x_B + \theta d_B$, and the equations decide by how much.

You choose only one number, how much of x_j to use. The equations decide everything else.

Proposition 5.3.2 (The j th basic direction). *Staying on $Ax = b$ forces $Ad = 0$, and hence*

$$d_B = -B^{-1}A_j.$$

This $d = d^{(j)}$ is called the j th basic direction. Its basic part is exactly the column of x_j in the tableau (Corollary 5.2.3(ii)).

Proof. $A(x + \theta d) = b$ and $Ax = b$ give $\theta Ad = 0$, so $Ad = 0$ because $\theta > 0$. All nonbasic entries of d vanish except $d_j = 1$, so

$$0 = Ad = \sum_{k=1}^m A_{B(k)} d_{B(k)} + A_j = Bd_B + A_j.$$

Multiply by B^{-1} . $\square$

Theorem 5.3.3 (Moving along a basic direction: the ratio test). *Let x be a BFS with basis B , let $j \in N$ and $d = d^{(j)}$. For $\theta \geq 0$:*

(a) $A(x + \theta d) = b$;

(b) $x + \theta d \geq 0$ if and only if $\theta \leq \theta_{\max}$, where

$$\theta_{\max} = \min_{k: d_{B(k)} < 0} \frac{x_{B(k)}}{-d_{B(k)}}, \quad \theta_{\max} = +\infty \text{ if } d_B \geq 0;$$

(c) $c^\top(x + \theta d) = c^\top x + \theta r_j$.

Consequently, if $r_j > 0$ and $d_B \geq 0$, the LP is **unbounded**. If $r_j > 0$ and $\theta_{\max} > 0$, moving to $x + \theta_{\max}d$ strictly increases the objective.

Proof. (a) $Ax = b$ and $Ad = 0$ (Proposition 5.3.2).

(b) Check each coordinate of $x + \theta d$.

- A nonbasic $i \neq j$: $0 + \theta \cdot 0 = 0$.
- The index j : $0 + \theta \geq 0$.
- A basic $B(k)$ with $d_{B(k)} \geq 0$: $x_{B(k)} + \theta d_{B(k)} \geq x_{B(k)} \geq 0$, so it imposes nothing.
- A basic $B(k)$ with $d_{B(k)} < 0$: $x_{B(k)} + \theta d_{B(k)} \geq 0$ if and only if $\theta \leq x_{B(k)}/(-d_{B(k)})$.

All coordinates are nonnegative exactly when θ satisfies every bound of the last kind, that is, when $\theta \leq \theta_{\max}$. If there is no such row, every $\theta \geq 0$ works.

(c) $c^\top d = c_j + c_B^\top d_B = c_j - c_B^\top B^{-1}A_j = r_j$.

Consequences. If $d_B \geq 0$, then by (a) and (b) $x + \theta d \in P$ for every $\theta \geq 0$, and by (c) its value $c^\top x + \theta r_j$ tends to $+\infty$ when $r_j > 0$. If $\theta_{\max} > 0$, the point $x + \theta_{\max}d$ is feasible and gains $\theta_{\max}r_j > 0$. $\square$

Key insight

Each basic row is a one-variable inequality $x_{B(k)} + \theta d_{B(k)} \geq 0$. The ratio test solves all of them and keeps the tightest. A zero ratio is allowed: if some $x_{B(k)} = 0$ with $d_{B(k)} < 0$, then $\theta_{\max} = 0$. This is degeneracy.

Theorem 5.3.4 (Basis exchange: the new point is a BFS). *In the setting of Theorem 5.3.3, suppose $\theta_{\max} < \infty$, and let ℓ be a row attaining the minimum, so $d_{B(\ell)} < 0$. Let B' be obtained from B by replacing the column $A_{B(\ell)}$ with A_j . Then B' is a basis, and $y := x + \theta_{\max}d$ is the BFS of B' .*

Proof. **Step 1 (B' is invertible).** Let $u := B^{-1}A_j = -d_B$. Since $B^{-1}A_{B(k)} = e_k$ for each k , the matrix $B^{-1}B'$ is the identity with its ℓ -th column replaced by u :

$$B^{-1}B' = [e_1 \cdots e_{\ell-1} \quad u \quad e_{\ell+1} \cdots e_m].$$

Row ℓ of this matrix is zero except for the entry u_ℓ on the diagonal, so expanding the determinant along row ℓ gives $\det(B^{-1}B') = u_\ell \cdot \det(I_{m-1}) = u_\ell = -d_{B(\ell)} > 0$. So $B^{-1}B'$ is invertible, and therefore so is $B' = B(B^{-1}B')$.

Step 2 (y is the basic solution of B'). The nonbasic indices of B' are $(N \setminus \{j\}) \cup \{B(\ell)\}$. For $i \in N \setminus \{j\}$, $y_i = 0$. Also $y_{B(\ell)} = x_{B(\ell)} + \theta_{\max}d_{B(\ell)} = 0$, by the choice of ℓ . So y is zero on every nonbasic index of B' , and $Ay = b$ by

Theorem 5.3.3(a). By Theorem 5.2.2(a) applied to B' , the only such vector is the basic solution of B' .

Step 3 (it is feasible). $y \geq 0$ by Theorem 5.3.3(b). $\square$

Interpretation & discussion

In tableau language this is the pivot of Chapter 3: x_j enters, $x_{B(\ell)}$ leaves, and the row of $x_{B(\ell)}$ is solved for x_j . That is possible because its coefficient $q_{\ell j} = d_{B(\ell)}^{(j)}$ is nonzero, which is exactly why B' is invertible.

Example 5.3.5 (Two basic directions at $(2, 2)$). Order the variables (x, y, s_1, s_2) , so $c = (3, 2, 0, 0)$. From Tableau 2, the basic directions are $d^{(s_1)} = (1, -2, 1, 0)$ and $d^{(s_2)} = (-1, 1, 0, 1)$.

j	$d_B^{(j)}$	$\theta_{\max}$	reaches	$r_j = c^\top d^{(j)}$	value there
s_1	$(1, -2)$	$2/2 = 1$ (y hits 0)	$(3, 0)$	$3 - 4 = -1$	$10 - 1 = 9$
s_2	$(-1, 1)$	$2/1 = 2$ (x hits 0)	$(0, 4)$	$-3 + 2 = -1$	$10 - 2 = 8$

Both moves lose 1 per unit, matching $z = 10 - s_1 - s_2$. As Theorem 5.3.4 predicts, moving along $d^{(s_1)}$ lands on the BFS of the basis $\{x, s_1\}$, which is Tableau 1, with y leaving.

5.4 Reduced Costs, Optimality and Termination

The gain along a basic direction

$$r_j = c^\top d^{(j)} = \underbrace{c_j}_{\text{direct gain}} - \underbrace{c_B^\top B^{-1} A_j}_{\text{cost of adjusting the basis}}.$$

Using one unit of x_j earns c_j directly, but forces the basic activities to shift, and that shift costs $c_B^\top B^{-1} A_j$.

Definition 5.4.1 (Reduced cost, multipliers, optimal basis). For a basis B :

- the *reduced costs* are $r^\top = c^\top - c_B^\top B^{-1} A$;
- the *multipliers* are $p^\top := c_B^\top B^{-1}$, so that $r_j = c_j - p^\top A_j$;
- B is an *optimal basis matrix* if $B^{-1}b \geq 0$ (feasibility) and $r \leq 0$ (optimality).

By Theorem 5.2.2, r_j is the coefficient of x_j in the objective row, and $r_B = 0$.

Theorem 5.4.2 (Reduced costs and optimality, maximisation). Let x be a BFS of $\max\{c^\top x : Ax = b, x \geq 0\}$ with basis matrix B .

1. If $r_j \leq 0$ for all $j \in N$, then x is optimal.
2. If x is optimal and nondegenerate, then $r_j \leq 0$ for all $j \in N$.

Proof. (1) Let u be any feasible plan. Theorem 5.2.2(b) applies to both u and x . Since $x_N = 0$:

$$c^\top u - c^\top x = \left(c_B^\top B^{-1} b + \sum_{j \in N} r_j u_j \right) - c_B^\top B^{-1} b = \sum_{j \in N} r_j u_j.$$

Each term is a nonpositive r_j times a nonnegative u_j , so the sum is ≤ 0 . Hence $c^\top u \leq c^\top x$ for every feasible u .

(2) Suppose, for contradiction, that $r_j > 0$ for some $j \in N$. Nondegeneracy means every $x_{B(k)} > 0$, so every ratio in Theorem 5.3.3(b) is positive and $\theta_{\max} > 0$. Then $x + \theta_{\max} d^{(j)}$ is feasible with value $c^\top x + \theta_{\max} r_j > c^\top x$, which contradicts the optimality of x . $\square$

Key insight

The identity $c^\top u - c^\top x = \sum_{j \in N} r_j u_j$ is the objective row of the tableau in matrix form, and it holds for every solution of $Au = b$. Part (1) only reads its signs. Part (2) is the contrapositive of “a positive reduced cost gives an improving move”, and that move needs room, which is what nondegeneracy provides.

Corollary 5.4.3 (Uniqueness). *If $r_j < 0$ for every $j \in N$, then x is the unique optimal solution.*

Proof. Let u be feasible with $c^\top u = c^\top x$. Then $\sum_{j \in N} r_j u_j = 0$, a sum of nonpositive terms, so every term is zero. Since every $r_j < 0$, this forces $u_j = 0$ for all $j \in N$. Then Theorem 5.2.2(a) gives $u_B = B^{-1}b = x_B$, so $u = x$. $\square$

Example 5.4.4 (Why part (2) needs nondegeneracy). Take $\max x_1$ subject to $x_1 + s = 0$, with $x_1, s \geq 0$. The only feasible point is $(0, 0)$, so it is optimal. The basis $\{s\}$ gives the degenerate BFS $s = 0$ and the tableau $s = 0 - x_1$, $z = 0 + x_1$, so $r_{x_1} = +1 > 0$. Yet no improvement is possible, because $\theta_{\max} = 0/1 = 0$. The basis $\{x_1\}$ describes the same point with $z = 0 - s$, which is a valid certificate. At a degenerate optimum, *some* basis certifies optimality, but not necessarily the one you are holding.

The simplex method, and why it stops

Theorem 5.4.5 (Termination without degeneracy). *Start from a feasible basis and repeat:*

- if $r_N \leq 0$, stop: the BFS is optimal;
- otherwise pick j with $r_j > 0$; if $d_B^{(j)} \geq 0$, stop: the LP is unbounded;
- otherwise exchange bases as in Theorem 5.3.4.

Each exchange changes the objective by $\theta_{\max} r_j \geq 0$. If every BFS met along the way is nondegenerate, the method stops after finitely many exchanges, with an optimal basis matrix or a proof of unboundedness.

Proof. The two stopping rules are correct by Theorem 5.4.2(1) and Theorem 5.3.3. By Theorem 5.3.3(c), an exchange changes the value by $\theta_{\max} r_j$, which is ≥ 0 . Under nondegeneracy $\theta_{\max} > 0$, so the value increases *strictly*. A basis determines its BFS, and hence its value. A strictly increasing sequence of values can never return to a basis already visited. There are at most $\binom{n}{m}$ bases, so the method stops. $\square$

Remark 5.4.6. Under degeneracy, $\theta_{\max} = 0$ is possible, the value can stall, and simplex can in principle cycle. Anti-cycling rules (for example Bland’s smallest-index rule) restore termination. Their proof is beyond this course; the theorems of this chapter never depend on them.

Optimal vertices exist

The sensitivity results about the optimal *value* need one more fact from Chapter 2: a feasible, bounded LP always has an optimal BFS.

Lemma 5.4.7 (Purification). *Suppose $c^\top x$ is bounded above on P . Then for every $x \in P$ there is a BFS x' with $c^\top x' \geq c^\top x$.*

If a feasible point is not a vertex, slide along a direction that keeps the equations and does not lose value, until a coordinate hits zero. Repeat until you reach a vertex.

Proof. By induction on the number of positive entries of x . Let $S = \{j : x_j > 0\}$.

Case 1: the columns A_S are linearly independent. Since $\text{rank } A = m$, these columns can be completed by further columns of A to m independent ones, which form a basis $B \supseteq S$. Then x is zero outside B and satisfies $Ax = b$. By Theorem 5.2.2(a), x is the basic solution of B , and it is feasible. Take $x' = x$.

Case 2: the columns A_S are dependent. Then there is $d \neq 0$ with $Ad = 0$ and $d_j = 0$ for $j \notin S$. Replacing d by $-d$ if necessary, $c^\top d \geq 0$. We may also assume that d has a negative entry:

- if $c^\top d > 0$ and $d \geq 0$, then $x + \theta d \in P$ for all $\theta \geq 0$ with value tending to $+\infty$, contradicting boundedness;
- if $c^\top d = 0$ and $d \geq 0$, replace d by $-d$.

Now let $\theta^* = \min\{x_j/(-d_j) : d_j < 0\}$. This is positive, because $d_j < 0$ only for $j \in S$, where $x_j > 0$. The point $x'' = x + \theta^* d$ satisfies:

- $Ax'' = b$;
- $x'' \geq 0$, by the same row-by-row argument as the ratio test;
- $c^\top x'' \geq c^\top x$;
- it has fewer positive entries than x , since no new positive entry appears outside S and the minimising index becomes 0.

By induction there is a BFS x' with $c^\top x' \geq c^\top x'' \geq c^\top x$. □

Theorem 5.4.8 (An optimal BFS exists). *If $P \neq \emptyset$ and $c^\top x$ is bounded above on P , the LP has an optimal solution that is a BFS. Moreover, its optimal value is the largest value $c^\top x$ among the finitely many BFSs.*

Proof. Applying Lemma 5.4.7 to any point of P shows that at least one BFS exists. There are finitely many bases, hence finitely many BFSs. Let x^* be a BFS of largest value. For any $x \in P$, the lemma gives a BFS x' with $c^\top x \leq c^\top x' \leq c^\top x^*$. So x^* is optimal. □

5.5 The Multipliers: Dual Solution and Shadow Prices

Duality, re-proved

Consider the canonical pair of Chapter 4:

$$(P) \quad \max c^\top x \quad \text{s.t.} \quad Ax \leq b, x \geq 0, \quad (D) \quad \min b^\top y \quad \text{s.t.} \quad A^\top y \geq c, y \geq 0.$$

In equational form, (P) adds slacks: $[A \ I](x, s) = b$, $(x, s) \geq 0$, with cost vector $(c, 0)$.

Theorem 5.5.1 (Weak duality). *For every feasible x of (P) and every feasible y of (D) , $c^\top x \leq b^\top y$.*

Proof. Insert the middle term $y^\top Ax$ and use one sign condition on each side:

$$c^\top x \leq (A^\top y)^\top x = y^\top Ax \leq y^\top b.$$

The first inequality holds because $A^\top y \geq c$ and $x \geq 0$. The second holds because $Ax \leq b$ and $y \geq 0$. Both use the same fact: if $u \leq v$ componentwise and $w \geq 0$, then $w^\top u \leq w^\top v$. $\square$

Theorem 5.5.2 (A matching pair is optimal). *If x is feasible for (P) , y is feasible for (D) , and $c^\top x = b^\top y$, then both are optimal.*

Proof. For every feasible x' , $c^\top x' \leq b^\top y = c^\top x$ by weak duality, so x is optimal. For every feasible y' , $b^\top y' \geq c^\top x = b^\top y$, so y is optimal. $\square$

Theorem 5.5.3 (The multipliers are an optimal dual solution). *Let B be an optimal basis matrix of the equational form of (P) , with BFS x^* . Then $p^\top = c_B^\top B^{-1}$ is optimal for (D) , and $b^\top p = c^\top x^*$.*

Proof. Step 1 (dual feasibility from $r \leq 0$). For an original column A_j , $r_j = c_j - p^\top A_j \leq 0$, so $(A^\top p)_j \geq c_j$; together, $A^\top p \geq c$. For the slack s_i , with column e_i and cost 0, $r_{s_i} = -p_i \leq 0$, so $p \geq 0$.

Step 2 (equal values). $b^\top p = c_B^\top B^{-1}b = c_B^\top x_B^* = c^\top x^*$, since $x_N^* = 0$ and slacks cost 0.

Step 3. Theorem 5.5.2. $\square$

This is Proposition 4.3.5 of Chapter 4 again, where $y^* = -r_s$. Indeed $r_s = -p$ by Corollary 5.2.3(iv). In the running example $p = (1, 1)$, with dual value $4 + 6 = 10$, equal to the primal optimum.

Corollary 5.5.4 (Complementary slackness at an optimal basis). *With B as in Theorem 5.5.3:*

- (a) *if the slack s_i is basic, then $p_i = 0$. Equivalently, $p_i > 0$ forces s_i nonbasic, so $s_i = 0$: the constraint is binding;*
- (b) *if x_j is basic, then $(A^\top p)_j = c_j$: the j -th dual constraint is tight.*

Proof. Basic reduced costs are zero (Theorem 5.2.2(c)). For s_i this reads $-p_i = 0$; for x_j it reads $c_j - (A^\top p)_j = 0$. $\square$

Interpretation & discussion

A resource with spare capacity is worth nothing at the margin. If s_i is basic, resource i is not fully used, and one more unit of it cannot help. Section 5.8 proves that p_i is exactly the value of one more unit of resource i , its *shadow price*, as long as the basis stays optimal.

Summary: Simplex in matrix form

- Every solution of $Ax = b$ satisfies $x_B = B^{-1}b - B^{-1}A_N x_N$ and $c^\top x = c_B^\top B^{-1}b + r_N^\top x_N$. The tableau is this formula, and it is unique.
- Basic direction: $d_B = -B^{-1}A_j$; step $\theta_{\max}$ by the ratio test; gain r_j per unit; the reached point is the BFS of the exchanged basis.
- $r_N \leq 0$ proves optimality, and strict signs prove uniqueness; the converse

needs nondegeneracy. Without degeneracy, simplex terminates.

- $p^\top = c_B^\top B^{-1}$ is the optimal dual solution; a basic slack has $p_i = 0$.

5.6 Sensitivity Analysis: The Framework

Motivation

A factory has its optimal plan. Then the questions arrive. Is one more hour of assembly worth buying, and at what price? A product's profit changes: is the plan still best? A new product is proposed: should we make it? Re-solving from scratch would work, but the final tableau already contains B^{-1} , p and the reduced costs. *Sensitivity analysis* reuses them to answer each question locally, around the known optimum.

We start from an optimal basis matrix B of $\max\{c^\top x : Ax = b, x \geq 0\}$, with optimum x^* ($x_B^* = B^{-1}b$, $x_N^* = 0$) and multipliers $p^\top = c_B^\top B^{-1}$.

Optimising once is rarely the end. The data move, and the question is whether yesterday's answer still holds.

Proposition 5.6.1 (Which test each change can break). (a) Changing b leaves B , B^{-1} , p and every reduced cost unchanged. Only the feasibility test $B^{-1}b \geq 0$ can fail.

(b) Changing c leaves $B^{-1}b$ unchanged, so x^* stays feasible. Only the optimality test $r \leq 0$ can fail.

(c) Adding a new variable with column a leaves B a basis, keeps x^* (extended by 0) as its BFS, and keeps the old reduced costs. It adds one new reduced cost.

In every case, B is optimal for the new data exactly when both tests hold for the new data (Theorem 5.4.2(1) and Definition 5.4.1).

Proof. (a) $r^\top = c^\top - c_B^\top B^{-1}A$ and $p^\top = c_B^\top B^{-1}$ do not involve b . (b) $B^{-1}b$ does not involve c . (c) Adding a column adds no equation, so B is still an invertible $m \times m$ matrix of columns of the new constraint matrix. Its basic solution is x^* with $u = 0$, and $r_j = c_j - p^\top A_j$ for the old columns is unchanged. $\square$

Change	What moves in the tableau	Test that can fail	If it fails
New variable (a, γ)	a new column	$r_u = \gamma - p^\top a \leq 0$	primal simplex pivot
$b \rightarrow b + \Delta b$	constants and z_0	$B^{-1}(b + \Delta b) \geq 0$	dual simplex pivot
$c \rightarrow c + \Delta c$	objective row only	$r' \leq 0$	primal simplex pivot

5.7 Case 1: A New Activity

Proposition 5.7.1 (Adding a variable). Add $u \geq 0$ with column $a \in \mathbb{R}^m$ and unit value γ . For the old basis B :

(a) $(x^*, 0)$ is still its BFS, and the new reduced cost is $r_u = \gamma - p^\top a$;

(b) if $r_u \leq 0$, then $(x^*, 0)$ is optimal for the enlarged LP;

(c) if $r_u > 0$ and x^* is nondegenerate, then $(x^*, 0)$ is not optimal, and u is

Price the new product at the shadow prices of the resources it would consume. Make it only if it earns more than that.

an improving entering variable. Its tableau column is $-B^{-1}a$.

Proof. (a) Proposition 5.6.1(c), plus $r_u = \gamma - c_B^\top B^{-1}a = \gamma - p^\top a$. (b) All reduced costs are now ≤ 0 ; apply Theorem 5.4.2(1). (c) Theorem 5.4.2(2) and Theorem 5.3.3. The column is Corollary 5.2.3(ii) with $A_j = a$. $\square$

Interpretation. $p^\top a$ is the value, at the current shadow prices, of the resources one unit of u would use. The new activity is worth considering only if it earns more than the resources it diverts, that is, $\gamma > p^\top a$.

Example 5.7.2 (A new activity using one unit of each resource). In the running example, $a = (1, 1)$ and $p = (1, 1)$, so $p^\top a = 2$. If $\gamma = 1.5$, then $r_u = -0.5$ and the old plan stays optimal. If $\gamma = 2.5$, then $r_u = +0.5$, and the basis loses its certificate.

Continue the old tableau. Append the column $-B^{-1}a = -(0, 1) = (0, -1)$ to the basic rows and $r_u = +\frac{1}{2}$ to the objective row:

$$x = 2 + s_1 - s_2 + 0u, \quad y = 2 - 2s_1 + s_2 - u, \quad z = 10 - s_1 - s_2 + \frac{1}{2}u.$$

u enters, and $y = 2 - u$ limits it to $u \leq 2$, so y leaves. Solving the y row for u and substituting:

$$x = 2 + s_1 - s_2, \quad u = 2 - 2s_1 + s_2 - y, \quad z = 11 - 2s_1 - \frac{1}{2}s_2 - \frac{1}{2}y.$$

All objective coefficients are ≤ 0 , so the new optimum is $(x, y, u) = (2, 0, 2)$ with $z = 11$. The new activity replaces y completely.

5.8 Case 2: A Resource Level Changes

Proposition 5.8.1 (Feasibility range and shadow prices). *Let $b' = b + \Delta b$, with A and c fixed.*

(a) *B is an optimal basis matrix for b' if and only if $B^{-1}(b + \Delta b) \geq 0$.*

(b) *In that case the optimum is $x'_B = B^{-1}(b + \Delta b)$, $x'_N = 0$, with value*

$$v(b + \Delta b) = c_B^\top B^{-1}(b + \Delta b) = v(b) + p^\top \Delta b.$$

So p_i is the shadow price of resource i : the change in the optimal value per extra unit of b_i , valid while B stays feasible.

Proof. (a) By Proposition 5.6.1(a), the reduced costs are unchanged and still ≤ 0 . So the only remaining condition in Definition 5.4.1 is $B^{-1}b' \geq 0$. (b) Theorem 5.4.2(1) gives optimality of the basic solution. Its value is $c_B^\top B^{-1}b' = c_B^\top B^{-1}b + c_B^\top B^{-1}\Delta b = v(b) + p^\top \Delta b$. $\square$

Proposition 5.8.2 (Ranging a right-hand side). *Let $\Delta b = \delta e_i$, and let $g = B^{-1}e_i$ be the i -th column of B^{-1} , which is minus the tableau column of s_i . Then B stays optimal exactly for $\delta_{\min} \leq \delta \leq \delta_{\max}$, where*

$$\delta_{\max} = \min_{k: g_k < 0} \frac{x_{B(k)}^*}{-g_k}, \quad \delta_{\min} = \max_{k: g_k > 0} \left(-\frac{x_{B(k)}^*}{g_k} \right),$$

with $\delta_{\max} = +\infty$ (or $\delta_{\min} = -\infty$) when no such k exists. The interval always

contains 0.

Proof. $B^{-1}(b + \delta e_i) = x_B^* + \delta g$, so by Proposition 5.8.1(a) the condition is $x_{B(k)}^* + \delta g_k \geq 0$ for every k . A row with $g_k = 0$ always holds. A row with $g_k < 0$ means $\delta \leq x_{B(k)}^*/(-g_k)$. A row with $g_k > 0$ means $\delta \geq -x_{B(k)}^*/g_k$. Intersecting all the rows gives the interval. It contains 0 because $x_B^* \geq 0$ makes every upper bound ≥ 0 and every lower bound ≤ 0 . $\square$

Example 5.8.3 (Resource 1 becomes $4 + \delta$). Here $g = B^{-1}e_1 = (-1, 2)$ and $x_B^* = (2, 2)$. The first row gives $\delta \leq 2/1 = 2$, and the second gives $\delta \geq -2/2 = -1$. So $\boxed{-1 \leq \delta \leq 2}$, and on that range $v(\delta) = 10 + p_1\delta = 10 + \delta$. In tableau form, only the constants and z_0 move:

$$x = (2 - \delta) + s_1 - s_2, \quad y = (2 + 2\delta) - 2s_1 + s_2, \quad z = (10 + \delta) - s_1 - s_2.$$

Beyond the range: the dual simplex step

Outside the range, some basic variable is negative, while every reduced cost is still ≤ 0 . The slides fix this with one pivot, choosing the *leaving* variable first. This is the *dual simplex* method.

The objective row is still a valid upper bound; only feasibility broke. So pivot to repair feasibility while keeping the objective row nonpositive.

Theorem 5.8.4 (The dual simplex step). Consider a tableau $x_{B(k)} = \beta_k + \sum_{j \in N} q_{kj}x_j$, $z = z_0 + \sum_{j \in N} r_jx_j$, with all $r_j \leq 0$.

- (a) **Upper bound.** Every feasible x has $c^\top x \leq z_0$. If moreover all $\beta_k \geq 0$, the basis is optimal.
- (b) **Infeasibility certificate.** If some $\beta_\ell < 0$ and $q_{\ell j} \leq 0$ for every $j \in N$, the LP is infeasible.
- (c) **The pivot.** If $\beta_\ell < 0$ and some $q_{\ell j} > 0$, let j^* minimise $(-r_j)/q_{\ell j}$ over the j with $q_{\ell j} > 0$, and pivot: x_{j^*} enters and $x_{B(\ell)}$ leaves. In the new tableau every reduced cost is still ≤ 0 , the constant of x_{j^*} is positive, and the new value z'_0 satisfies $z'_0 \leq z_0$, strictly if $r_{j^*} < 0$.

Proof. (a) The objective row holds for every solution of $Ax = b$, by Theorem 5.2.2(b). For a feasible x , each $r_jx_j \leq 0$, so $c^\top x \leq z_0$. If also $\beta \geq 0$, the basis passes both tests.

(b) Every solution of $Ax = b$ satisfies the row $x_{B(\ell)} = \beta_\ell + \sum_j q_{\ell j}x_j$. If $x \geq 0$, each term $q_{\ell j}x_j$ is ≤ 0 , so $x_{B(\ell)} \leq \beta_\ell < 0$. This contradicts $x_{B(\ell)} \geq 0$, so no feasible x exists.

(c) Write $q := q_{\ell j^*} > 0$ and $r := r_{j^*} \leq 0$. The new basis is valid, since $q \neq 0$ (Theorem 5.3.4, Step 1). Solving row ℓ for x_{j^*} gives

$$x_{j^*} = -\frac{\beta_\ell}{q} + \frac{1}{q}x_{B(\ell)} - \sum_{k \neq j^*} \frac{q_{\ell k}}{q}x_k,$$

whose constant $-\beta_\ell/q$ is positive. Substituting into the objective row:

$$z = \left(z_0 - \frac{r\beta_\ell}{q}\right) + \frac{r}{q}x_{B(\ell)} + \sum_{k \in N, k \neq j^*} \left(r_k - \frac{r q_{\ell k}}{q}\right)x_k.$$

- **New constant:** $-r\beta_\ell/q = (-r)\beta_\ell/q \leq 0$, because $-r \geq 0$, $\beta_\ell < 0$ and $q > 0$. So $z'_0 \leq z_0$, strictly if $r < 0$.

- *Coefficient of $x_{B(\ell)}$* : $r/q \leq 0$.
- *Coefficient of x_k with $q_{\ell k} > 0$* : it equals $q_{\ell k}[(-r)/q - (-r_k)/q_{\ell k}]$, which is ≤ 0 because j^* minimised the ratio.
- *Coefficient of x_k with $q_{\ell k} \leq 0$* : it equals $r_k + (-r)q_{\ell k}/q$, a sum of two terms that are each ≤ 0 . $\square$

Interpretation & discussion

The primal simplex keeps *feasibility* and climbs the value until the reduced costs are nonpositive. The dual simplex keeps *nonpositive reduced costs*, so z_0 is always an upper bound by (a), and lowers that bound until feasibility is restored. If every pivot has $r_{j^*} < 0$, the bound strictly decreases, no basis repeats, and the method stops, exactly as in Theorem 5.4.5. This is the natural tool after a change of b , because such a change never breaks the reduced costs.

Example 5.8.5 (The whole value curve). $\delta > 2$: $x = 2 - \delta < 0$. In its row, only s_1 has a positive coefficient, so s_1 enters and x leaves:

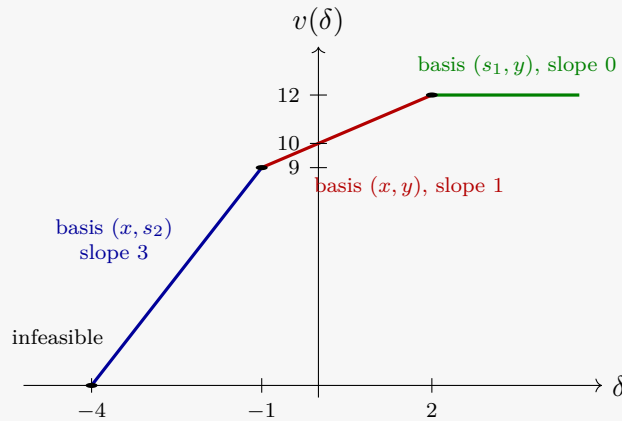
$$s_1 = (\delta - 2) + x + s_2, \quad y = 6 - 2x - s_2, \quad z = 12 - x - 2s_2.$$

This is feasible for $\delta \geq 2$, with $v = 12$. The bound fell from $10 + \delta > 12$ to 12, as part (c) predicts. Since s_1 is now basic, $p_1 = 0$ (Corollary 5.5.4): extra units of resource 1 are worthless.

$\delta < -1$: $y = 2 + 2\delta < 0$. Only s_2 has a positive coefficient in its row, so s_2 enters and y leaves:

$$s_2 = (-2 - 2\delta) + y + 2s_1, \quad x = (4 + \delta) - y - s_1, \quad z = (12 + 3\delta) - y - 3s_1.$$

This is feasible for $-4 \leq \delta \leq -1$, with $v = 3(4 + \delta)$. For $\delta < -4$, the row $x = (4 + \delta) - y - s_1$ has a negative constant and no positive coefficient, so by part (b) the LP is infeasible.



Theorem 5.8.6 (The optimal value as a function of b). Let $v(b)$ denote the optimal value of $\max\{c^\top x : Ax = b, x \geq 0\}$ when it is finite.

- (a) **Concavity.** If $v(b_1)$ and $v(b_2)$ are finite and $0 \leq \lambda \leq 1$, then the LP at $b_\lambda = \lambda b_1 + (1 - \lambda)b_2$ is feasible, and $v(b_\lambda) \geq \lambda v(b_1) + (1 - \lambda)v(b_2)$ whenever $v(b_\lambda)$ is finite.

(b) **Finitely many linear pieces.** Whenever $v(b)$ is finite,

$$v(b) = \max\{c_B^\top B^{-1}b : B \text{ a basis with } B^{-1}b \geq 0\}.$$

So at every such b , v coincides with one of the finitely many linear functions $b \mapsto p_B^\top b$, where $p_B^\top = c_B^\top B^{-1}$.

(c) **Slopes are shadow prices.** On an interval of δ where a single basis B stays optimal for $b + \delta e_i$, v is linear with slope $(p_B)_i$.

Proof. (a) By Theorem 5.4.8 there are optimal x_1, x_2 for b_1, b_2 . Let $x_\lambda = \lambda x_1 + (1 - \lambda)x_2$. It is ≥ 0 , and $Ax_\lambda = \lambda b_1 + (1 - \lambda)b_2 = b_\lambda$, so it is feasible for b_λ . Its value is $\lambda v(b_1) + (1 - \lambda)v(b_2)$, and the optimum at b_λ is at least that.

(b) Each feasible basis gives a feasible point of value $c_B^\top B^{-1}b$, so the maximum is $\leq v(b)$. By Theorem 5.4.8 some BFS is optimal, so the maximum is attained and equals $v(b)$. There are finitely many bases.

(c) Proposition 5.8.1(b) with $\Delta b = \delta e_i$. $\square$

Interpretation & discussion

A concave function made of linear pieces has *decreasing* slopes. In the example the slopes are $3 \rightarrow 1 \rightarrow 0$: each extra unit of resource 1 is worth less as it becomes more plentiful. This is the economics of diminishing returns, and here it is a theorem. A shadow price is a *local* statement, valid only on its range.

5.9 Case 3: A Profit Changes

Proposition 5.9.1 (Changing the objective). Let $c' = c + \Delta c$, with A and b fixed, and write the tableau rows as $x_{B(k)} = \beta_k + \sum_{j \in N} q_{kj}x_j$.

- (a) x^* stays feasible, and the new reduced costs are $(r')^\top = r^\top + \Delta c^\top - \Delta c_B^\top B^{-1}A$.
- (b) **Nonbasic change.** If only c_j ($j \in N$) changes, by t , then $r'_j = r_j + t$ and every other reduced cost is unchanged. B stays optimal if and only if $t \leq -r_j$.
- (c) **Basic change.** If only $c_{B(k)}$ changes, by ε , then $z'_0 = z_0 + \varepsilon\beta_k$ and $r'_j = r_j + \varepsilon q_{kj}$ for $j \in N$. In words: add ε times the row of $x_{B(k)}$ to the objective row.

Proof. (a) Feasibility is Proposition 5.6.1(b). For the reduced costs, expand $(c + \Delta c)^\top - (c_B + \Delta c_B)^\top B^{-1}A$ and group the terms containing c into $r^\top$.

(b) $\Delta c_B = 0$, so $r' = r + \Delta c$, which changes only entry j . Then $r' \leq 0$ holds exactly when $r_j + t \leq 0$.

(c) For every solution of $Ax = b$, the new objective is $(c')^\top x = c^\top x + \varepsilon x_{B(k)}$. Substitute the objective row and the row of $x_{B(k)}$, both valid for all solutions (Theorem 5.2.2):

$$(c')^\top x = \left(z_0 + \sum_{j \in N} r_j x_j\right) + \varepsilon \left(\beta_k + \sum_{j \in N} q_{kj} x_j\right) = (z_0 + \varepsilon\beta_k) + \sum_{j \in N} (r_j + \varepsilon q_{kj}) x_j.$$

By the uniqueness in Theorem 5.2.2(d), applied to c' , these are the new constant and reduced costs. $\square$

Corollary 5.9.2 (Ranging a basic profit). *In case (c), B stays optimal exactly for $\varepsilon_{\min} \leq \varepsilon \leq \varepsilon_{\max}$, where*

$$\varepsilon_{\max} = \min_{j: q_{kj} > 0} \frac{-r_j}{q_{kj}}, \quad \varepsilon_{\min} = \max_{j: q_{kj} < 0} \frac{-r_j}{q_{kj}}.$$

Since $r_j \leq 0$, $\varepsilon_{\max} \geq 0 \geq \varepsilon_{\min}$. The plan x^ stays optimal on this interval, with value $z_0 + \varepsilon\beta_k$.*

Proof. Require $r_j + \varepsilon q_{kj} \leq 0$ for each $j \in N$. For $q_{kj} > 0$ this gives $\varepsilon \leq -r_j/q_{kj}$, and for $q_{kj} < 0$ it gives $\varepsilon \geq -r_j/q_{kj}$; the case $q_{kj} = 0$ always holds. Intersect. $\square$

Example 5.9.3 (The profit of x becomes $3+\varepsilon$). The row of x is $x = 2 + s_1 - s_2$, so $q = (1, -1)$ on (s_1, s_2) , while $r = (-1, -1)$. Then $\varepsilon_{\max} = 1/1 = 1$ and $\varepsilon_{\min} = 1/(-1) = -1$. Adding $\varepsilon \times$ the row:

$$z' = (10 + 2\varepsilon) + (\varepsilon - 1)s_1 - (1 + \varepsilon)s_2,$$

so $[-1 \leq \varepsilon \leq 1]$, with value $10 + 2\varepsilon$. This agrees with $p^\top = (3 + \varepsilon, 2)B^{-1} = (1 - \varepsilon, 1 + \varepsilon)$.

At $\varepsilon = 2$: $z' = 14 + s_1 - 3s_2$. Now s_1 enters, and $y = 2 - 2s_1$ limits it to $s_1 \leq 1$, so y leaves:

$$x = 3 - \frac{1}{2}y - \frac{1}{2}s_2, \quad s_1 = 1 - \frac{1}{2}y + \frac{1}{2}s_2, \quad z' = 15 - \frac{1}{2}y - \frac{5}{2}s_2.$$

The new optimum is $(3, 0)$, with value 15.

Interpretation & discussion

Geometry. Both x and y are basic, so Corollary 5.5.4(b) gives $c' = A^\top p'$, that is $(3 + \varepsilon, 2) = p'_1(1, 1) + p'_2(2, 1)$. This writes the objective as a combination of the normals of the two binding constraints, with weights p' . Optimality of the basis is $p' \geq 0$, so the vertex $(2, 2)$ is optimal exactly when the objective vector lies in the cone spanned by those normals. Solving gives $p' = (1 - \varepsilon, 1 + \varepsilon)$, nonnegative for $-1 \leq \varepsilon \leq 1$. At $\varepsilon = \pm 1$ the iso-profit line lies along an edge, and two vertices are optimal.

Proposition 5.9.4 (The optimal value as a function of c). *Fix A , b with $P \neq \emptyset$, and let $w(c)$ be the optimal value when finite. Then $w(c) = \max\{c^\top x : x \text{ a BFS}\}$. So w is a maximum of finitely many linear functions of c : it is **convex** and piecewise linear. On a range where one BFS x^* stays optimal, $w(c + \varepsilon e_k) = w(c) + \varepsilon x_k^*$.*

Proof. The formula is Theorem 5.4.8. A maximum of linear functions f_i is convex: $\max_i f_i(\lambda c + (1 - \lambda)c') \leq \lambda \max_i f_i(c) + (1 - \lambda) \max_i f_i(c')$, because each f_i is linear. The last claim is $(c + \varepsilon e_k)^\top x^* = c^\top x^* + \varepsilon x_k^*$. $\square$

In the example, $w = 10 + 2\varepsilon$ (slope $x^* = 2$) up to $\varepsilon = 1$, then $w = 3(3 + \varepsilon) = 9 + 3\varepsilon$ (slope $x^* = 3$, value 15 at $\varepsilon = 2$). The slopes increase, as convexity requires. The value is concave in b but convex in c .

5.10 A Complete Example: Production Planning

Example 5.10.1 (The workshop's three computers). Models A, B and C use assembly time and memory boards:

	A	B	C	Available
Assembly time	1	1	1	8
Memory boards	2	1	1	10
Profit per unit	4	3	2	

$$\begin{aligned} \max \quad & 4x_A + 3x_B + 2x_C \\ \text{s.t.} \quad & x_A + x_B + x_C + s_1 = 8, \quad 2x_A + x_B + x_C + s_2 = 10, \quad x, s \geq 0. \end{aligned}$$

Solve once. The optimal basis is (x_A, x_B) , with

$$B = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}, \quad B^{-1}b = \begin{bmatrix} 2 \\ 6 \end{bmatrix} \geq 0,$$

and $p^\top = (4, 3)B^{-1} = (2, 1)$. The reduced costs are $r_C = 2 - (2 + 1) = -1$, $r_{s_1} = -2$ and $r_{s_2} = -1$, all < 0 . So by Corollary 5.4.3, $(x_A, x_B, x_C) = (2, 6, 0)$ is the unique optimum, with profit 26. Both resources are used up ($2 + 6 = 8$, $4 + 6 = 10$), consistent with $p > 0$ (Corollary 5.5.4). The final tableau, to be reused, is

$$x_A = 2 + 0x_C + s_1 - s_2, \quad x_B = 6 - x_C - 2s_1 + s_2, \quad z = 26 - x_C - 2s_1 - s_2.$$

The shadow prices are $p = (2, 1)$: one more assembly unit is worth 2, and one more board is worth 1, locally. Each scenario below changes the *original* model independently.

Scenario	Test	Verdict	New optimum
New model D: $a_D = (1, 1)$, $\gamma_D = 3.5$	$r_D = 3.5 - (2, 1) \cdot (1, 1) = +0.5$	basis fails; x_D enters, x_B leaves ($x_D \leq 6$)	$(x_A, x_D) = (2, 6)$, $z = 29$
Assembly $8 + \delta$	$B^{-1}(8 + \delta, 10) = (2 - \delta, 6 + 2\delta) \geq 0$	basis kept for $-3 \leq \delta \leq 2$	$z = 26 + 2\delta$; at $\delta = 1$: $(1, 8, 0)$, 28
Profit of A: $4 + \varepsilon$	row of x_A added: $r = (-1, \varepsilon - 2, -1 - \varepsilon)$	basis kept for $-1 \leq \varepsilon \leq 2$	$(2, 6, 0)$, $z = 26 + 2\varepsilon$

For model D, the pivot gives $z = 29 - \frac{1}{2}x_B - \frac{3}{2}x_C - 3s_1 - \frac{1}{2}s_2$, so the certificate is immediate. All three scenarios, and the extra ones below, were checked by solving the modified LPs exactly.

5.11 Techniques, Classic Problems and Gotchas

Technique: Exam-speed sensitivity

1. From the final tableau, read B^{-1} as *minus* the slack columns, p as *minus* the slack reduced costs, and x_B as the constants.
2. **New activity:** $r_u = \gamma - p^\top a$. Its tableau column is $-B^{-1}a$.

3. **RHS change** δe_i : new constants $x_B + \delta B^{-1}e_i \geq 0$ give the range; the value is $v + p_i\delta$ inside it.
4. **Nonbasic profit change** t : keep the basis while $t \leq -r_j$.
5. **Basic profit change** ε : add $\varepsilon \times (\text{its row})$ to the objective row; keep the basis while all coefficients are ≤ 0 .
6. **Outside a range**: an infeasible constant calls for a dual simplex pivot; a positive objective coefficient calls for an ordinary pivot.

Technique: Theory checklist: what to state and how to prove it

- **Tableau of a basis** (Thm 5.2.2): split $Ax = Bx_B + A_Nx_N$, solve for x_B , substitute into $c^\top x$. Uniqueness: plug $x_N = 0$, then $x_N = e_j$.
- **Basic direction** (Prop 5.3.2): $Ad = 0$ with $d_j = 1$ gives $Bd_B + A_j = 0$.
- **Ratio test** (Thm 5.3.3): check coordinates one by one; only rows with $d_{B(k)} < 0$ bound θ . Gain: $c^\top d = r_j$.
- **Exchange** (Thm 5.3.4): $B^{-1}B'$ is I with one column replaced by u , so $\det = u_\ell \neq 0$; the new point vanishes on the new nonbasics.
- **Optimality** (Thm 5.4.2): $c^\top u - c^\top x = \sum_N r_j u_j$. Converse: an improving step exists when $\theta_{\max} > 0$. **Uniqueness**: all terms zero with $r_j < 0$.
- **Termination** (Thm 5.4.5): strict increase, finitely many bases. **Existence** (Thm 5.4.8): purification by induction on the support.
- p is **dual-optimal** (Thm 5.5.3): $r \leq 0$ on original columns gives $A^\top p \geq c$, and on slacks gives $p \geq 0$; values match; weak duality inserts $y^\top Ax$.
- **Shadow prices and ranges** (Props 5.8.1, 5.8.2): b does not enter r ; value $c_B^\top B^{-1}(b + \Delta b)$; one inequality per row.
- **Dual simplex** (Thm 5.8.4): z_0 is an upper bound; the minimum-ratio rule keeps $r \leq 0$; a row with no positive coefficient proves infeasibility.
- **Value functions** (Thm 5.8.6, Prop 5.9.4): concave in b by mixing optimal plans; convex in c as a maximum over vertices.

Classic problems

- *Production example: boards become $10 + \delta$. Range and value?* $g = B^{-1}e_2 = (1, -1)$, so $2 + \delta \geq 0$ and $6 - \delta \geq 0$: $-2 \leq \delta \leq 6$, with $z = 26 + \delta$ (slope $p_2 = 1$). At $\delta = 6$: $(8, 0, 0)$, 32.
- *Assembly becomes 11 ($\delta = 3$, beyond the range)?* $x_A = -1 + s_1 - s_2$ is infeasible, and only s_1 has a positive coefficient, so s_1 enters and x_A leaves: $x_B = 10 - 2x_A - x_C - s_2$, $z = 30 - 2x_A - x_C - 3s_2$. The optimum is $(0, 10, 0)$ with 30. Now s_1 is basic, so $p_1 = 0$: extra assembly time is worth nothing.
- *A product E with $a = (1, 2)$ earning 3?* $r_E = 3 - (2 \cdot 1 + 1 \cdot 2) = -1$: do not make it. It breaks even at a profit of 4.
- *How profitable must model C become before it is worth making?* $r_C = -1$, so by Proposition 5.9.1(b) its profit must exceed $2 + 1 = 3$.
- *Running example: shadow price and range of resource 2?* $p_2 = 1$, and $g = B^{-1}e_2 = (1, -1)$, so $2 + \delta \geq 0$ and $2 - \delta \geq 0$: $-2 \leq \delta \leq 2$.

- *Prove: if the optimal basis has every reduced cost strictly negative, a small change of any one c_j does not change the optimal plan.* By Corollary 5.9.2 and Proposition 5.9.1(b), each range has $\varepsilon_{\max} > 0 > \varepsilon_{\min}$, because every $-r_j > 0$.

Gotcha

- The tableau shows $-B^{-1}$ in the slack columns, and $-p$ as the slack reduced costs. Two minus signs, both easy to lose.
- These are **maximisation** conventions: $r_j > 0$ means improving. Many textbooks (for example Bertsimas–Tsitsiklis) minimise, and there $\bar{c}_j < 0$ means improving. Translate before comparing.
- A shadow price holds only on its range. Beyond it, the basis changes and so does the slope.
- A change in a *basic* profit moves p , so every reduced cost must be rechecked, not just one.
- Changing b never breaks the reduced costs; changing c never breaks feasibility. Know which test each change can affect (Proposition 5.6.1).
- Part (2) of Theorem 5.4.2 fails at degenerate points (Example 5.4.4).
- The value is *concave* in b and *convex* in c ; do not mix them up.

Notes on the original slides

Every tableau, range and pivot on the Lecture 5 slides was recomputed, both by hand and by solving each modified LP exactly. All are correct. Points worth knowing:

- **Slides 10 and 20** make the same point (B^{-1} from the slack columns) twice. Corollary 5.2.3 states it once, in general, and Theorem 5.2.2 proves why it holds.
- **Slides 34–35** repair infeasibility by choosing the leaving variable first, the negative basic one. This is the *dual simplex* method, which the slides use without naming. Theorem 5.8.4 gives the entering rule and proves it.
- **Slide 15**, part 2, needs nondegeneracy, and Example 5.4.4 shows why. The slides prove it but give no counterexample.
- **Slide 24** introduces $p^\top = c_B^\top B^{-1}$ as “resource multipliers”. These are exactly the optimal dual variables of Chapter 4 (Theorem 5.5.3).
- **Slide 36**: the plot’s vertical axis marks 9 and 12. The curve also passes through (0, 10) and reaches 0 at $\delta = -4$. Its concavity is Theorem 5.8.6.

Chapter takeaways

Summary: Simplex tableau and sensitivity

- The tableau of a basis is the unique formula $x_B = B^{-1}b - B^{-1}A_Nx_N$, $z = c_B^\top B^{-1}b + r_N^\top x_N$. Everything else is read from it.
- A simplex move follows $d_B = -B^{-1}A_j$, gains r_j per unit, and stops at the ratio-test bound on a new BFS. B is optimal when $B^{-1}b \geq 0$ and $r \leq 0$; then $p^\top = c_B^\top B^{-1}$ is dual-optimal.
- New activity: $r_u = \gamma - p^\top a$. New b : keep B while $B^{-1}(b + \Delta b) \geq 0$, with value $v + p^\top \Delta b$. New c : keep B while $r' \leq 0$.
- Outside a range, pivot from the saved tableau: dual simplex for a negative constant, primal for a positive r_j . The value is concave piecewise linear in b , and convex in c .